

Lagra as a measurement harness for pde-engine force-free validation

Pim de Witte · Lagra Project · pde-engine adapter note · May 26, 2026

Abstract

This note rewrites the pde-engine force-free adapter in the visual style of *Lagra: A differentiable physics kernel for verifiable domain operations*. The purpose is narrower than the Lagra kernel paper. We do not claim a new literature-priority result or a full reproduction of the pde-engine discovery pipeline. We show how Lagra enters a local pde-engine checkout, which inputs are admitted into the Lagra proof-search graph, which quantities are measured, and which claims are allowed into the paper. The positive result is exact and useful: seven pde-engine registry force-free foliations simplify to determinant zero; two negative controls simplify to nonzero determinants; three rational checkpoints separate known solutions from controls; and a rotation-context cache regression proves that the validator cache key must include Ω and the validation mode; and a bounded engine pass exports six non-registered, independently rechecked candidate foliations as a process-control baseline rather than as a publication target. A second gate records the next Kerr paper target, appends targeted one-term and two-term finite-spin correction grammars, runs a 48-coefficient leading-order solve, checks the literature slow-rotation perturbative anchor, and emits `no_candidate_yet` rather than promoting the current linear surrogate. The negative result is equally important: the bounded pde-engine smoke is now process-clean, but the full seven-solution pde-engine/Lean reproduction path remains non-green on this checkout and is recorded as a boundary, not upgraded into a proof.

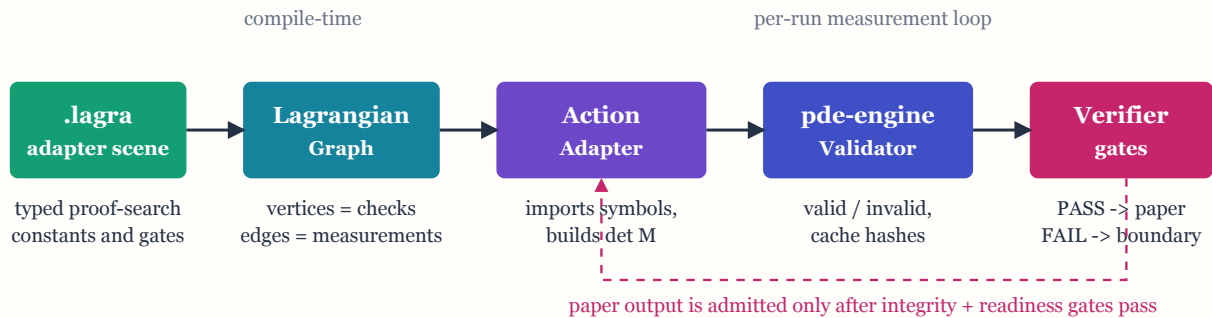
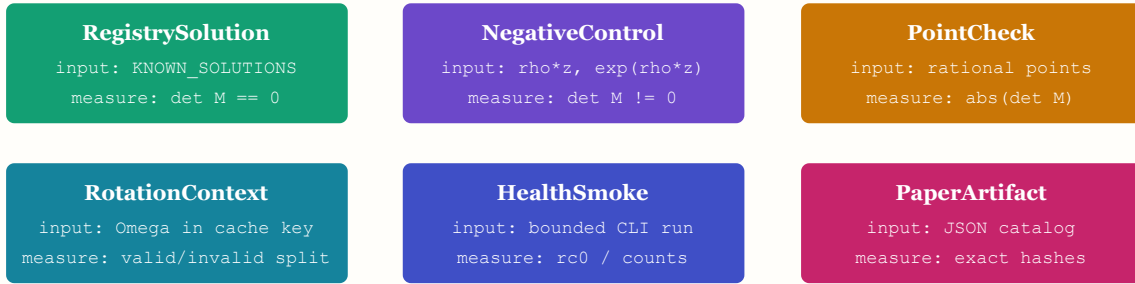


Figure 1: The Lagra/pde-engine adapter pipeline in one figure. This is the direct analogue of the Lagra pipeline figure. The input is a typed Lagra proof-search source plus a local pde-engine checkout. The measured output is not a trajectory; it is a JSON evidence artifact and an admitted paper section. The dashed return path is the strict gate: failed measurements do not become prose.

```
AdapterKind = RegistrySolution | NegativeControl | PointCheck
              | RotationContext | HealthSmoke | PaperArtifact
```



Every chip has a typed input, a deterministic measurement, and an admitted output field.

Figure 2: The adapter vocabulary. The original Lagra vocabulary figure lists LagrangianNode variants. Here the closed vocabulary is the set of adapter checks. Each variant names exactly what enters the check and exactly what is measured on the way out.

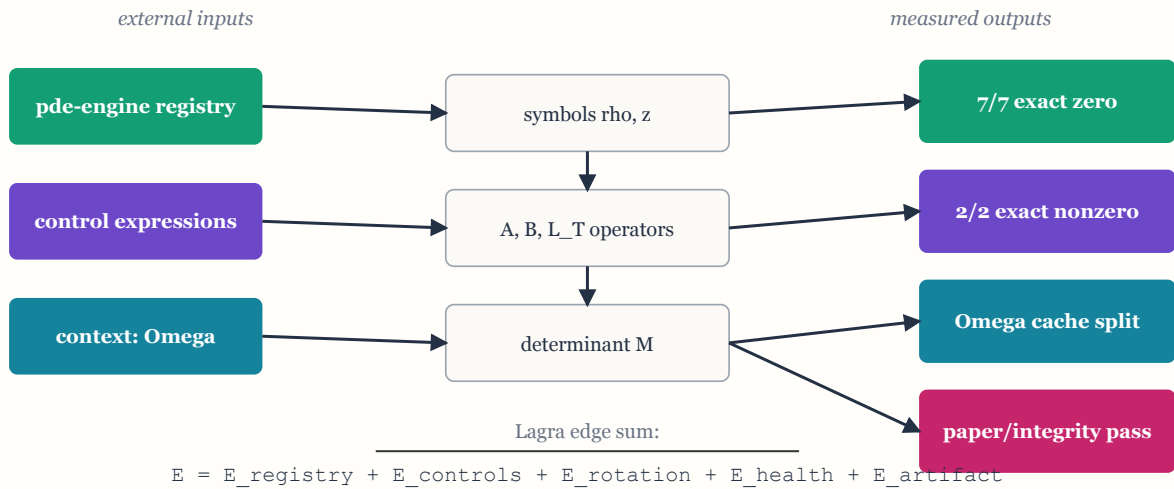


Figure 3: Anatomy of the Lagra adapter graph. The external pde-engine registry, control expressions, and validation context enter as graph-visible inputs. The edges are measurement obligations. The outputs are the counts and facts consumed by the proof-search catalog: exact-zero solutions, exact-nonzero controls, rotation-cache separation, and paper-readiness status.

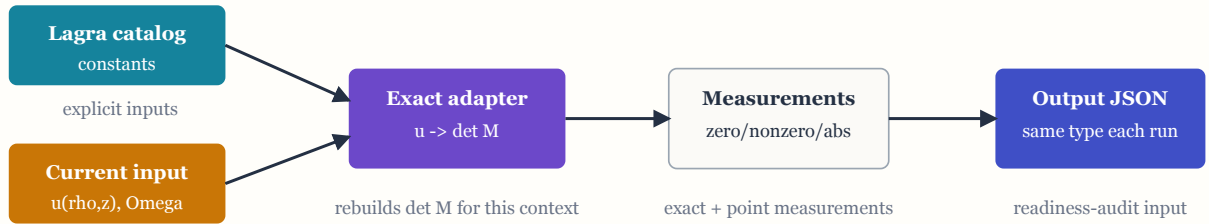


Figure 4: One adapter run as a data-flow diagram. This mirrors the one-timestep figure in the Lagra paper. The action is not a mechanical time step; it is one deterministic validation pass. The adapter rebuilds the symbolic determinant for the current expression and context, measures exact and pointwise quantities, and writes a typed JSON artifact.

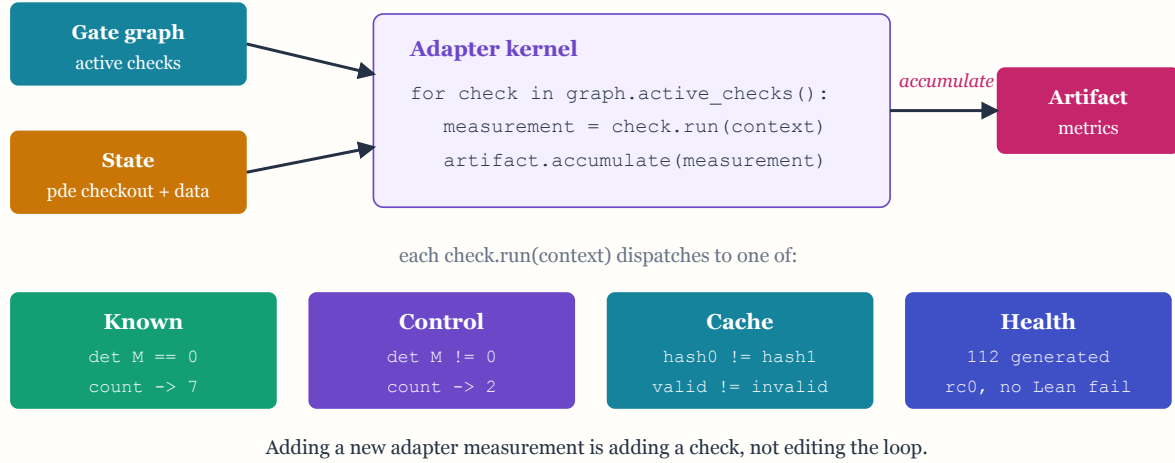
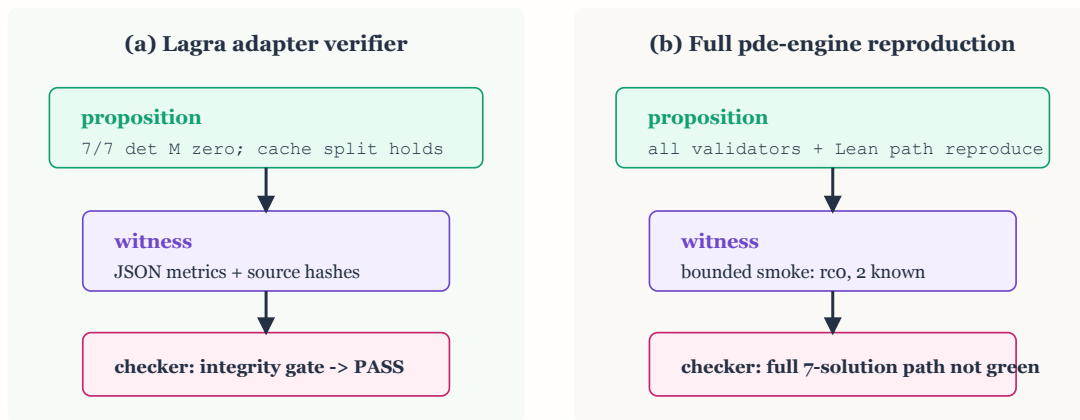


Figure 5: The adapter kernel, exploded. The loop is deliberately boring. Heterogeneity lives in `check.run(context)`, not in the loop. Adding a Kerr no-promotion guard, a force-free cache regression, or a future PDE boundary check means adding a new typed check with a measurement contract.



Same discipline as the Lagra verifier triangle; different claim admitted.

Figure 6: Verifier triangle for the adapter and the full reproduction boundary. This copies the structural shape of the Lagra verifier triangle while keeping the trust event honest. The adapter proposition passes with JSON witnesses and source hashes. The bounded pde-engine smoke is now process-clean: return code 0, no Lean build failure, 112 generated expressions, 76 valid rows, and 2 known vertical forms. The full seven-solution reproduction proposition still remains non-green because the bounded depth-2 smoke does not recover all seven known solutions.

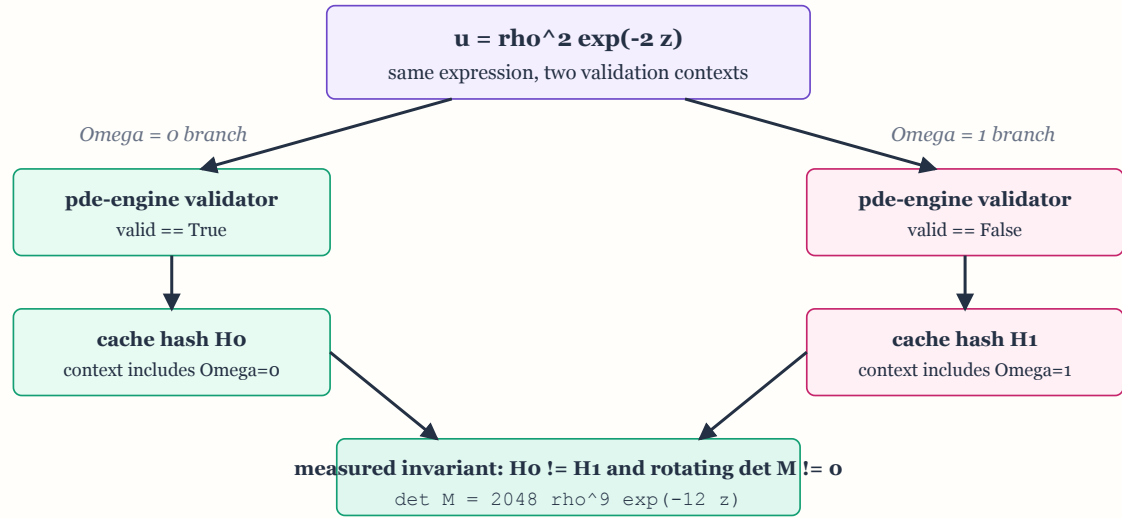
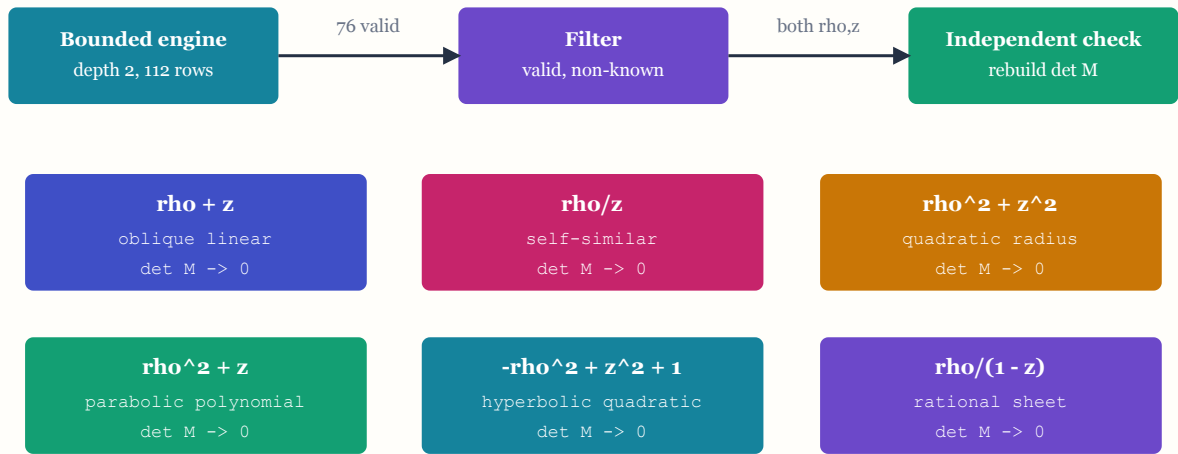


Figure 7: Verifiability by derivation for the cache regression. The same expression is sent down two independent validation-context branches. Lagra measures the branch split, the pde-engine validator result, and the exact rotating determinant. The conclusion is a code obligation: the validator cache key must include Omega and validation mode.



Novelty is scoped to the repo registry: not identical to the seven registered known_solutions.

Figure 8: Engine-discovered candidate foliations admitted by the adapter. The bounded depth-2 pde-engine run produces valid rows that are not identical to the seven registered known solutions. The exporter does not trust the cache: it rebuilds the determinant independently and admits only both-variable candidates whose determinant simplifies exactly to zero. This is an engine-novel result relative to the repository registry, not a literature-priority claim.

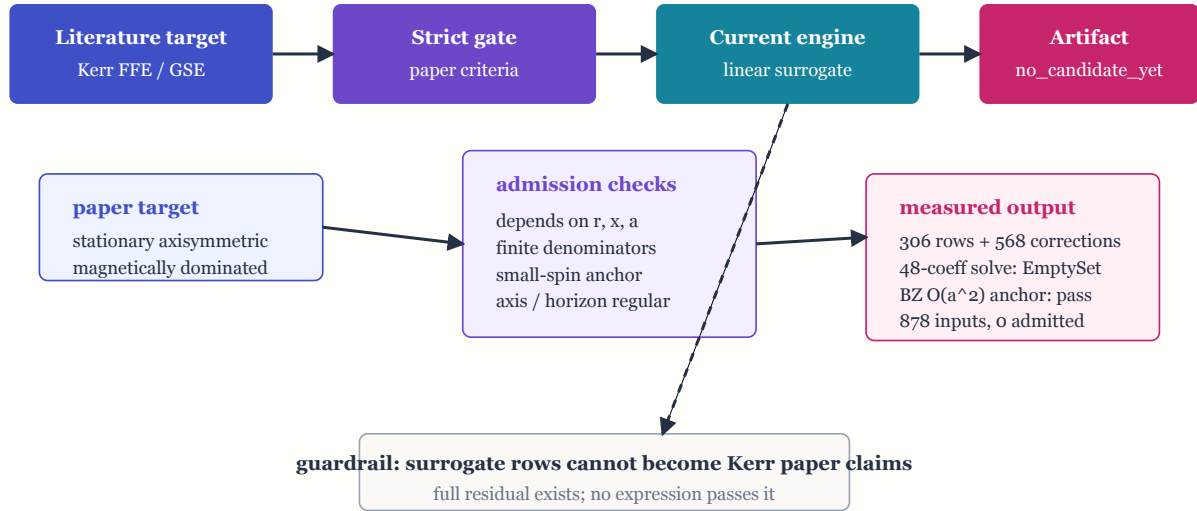


Figure 9: Kerr paper-target gate. The next target uses the same measurement discipline in the negative direction. The bounded Kerr run generated 306 rows; the strict gate scans those rows, 456 targeted finite-spin corrections, 112 Kerr-metric-resummed corrections, and four probes against the closed split-monopole nonlinear Kerr Grad-Shafranov residual. It also solves the 48-coefficient leading-order ansatz and gets EmptySet, verifies the known $O(a^2)$ Blandford-Znajek perturbative anchor, then emits `no_candidate_yet` because no finite-spin exact expression passes the criteria.

1. What exactly enters Lagra

The adapter has four external inputs. First, it imports the pde-engine force-free symbols `rho` and `z` from `problems/force_free/validator.py`. Second, it loads the seven known Compere force-free foliations from the pde-engine problem registry. Third, it supplies two false controls, `rho*z` and `exp(rho*z)`. Fourth, it runs one rotation-context check on `rho**2*exp(-2*z)` with $\Omega=0$ and $\Omega=1$ against a shared cache database.

Nothing enters through prose. Each input is bound into a Lagra proof-search artifact and then checked by `publication_readiness_audit.py` and `proof_search_integrity_gate.py`. The paper is downstream of those gates. This is the central point of using Lagra here: the artifact catalog, not the narrative, decides whether the claim is admissible.

2. What exactly is measured

For an expression $u(\rho, z)$, the adapter evaluates the force-free determinant from Compere, Gralla, and Lupsasca, *Force-Free Foliations*, Phys. Rev. D 94, 124012 (2016), Eq. 2.14 / Section 2.4:

```
det([[L_T(A), L_T(B)], [L_T^2(A), L_T^2(B)]])
```

```
A = u_rho_rho + u_z_z - u_rho/rho
B = u_rho**2 + u_z**2
T = u_z*d_rho - u_rho*d_z
```

The strict measurements are:

Measurement	Observed result	Claim admitted
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Registry exact determinant	7/7 simplify to 0	The Lagra adapter matches the pde-engine registry solutions for this determinant.
Negative-control exact determinant	$\rho * z \rightarrow 16 * \rho * z; \exp(\rho * z) \rightarrow 16 * \rho * z * \exp(6 * \rho * z)$	The operator is not vacuously zero.
Rational point checks	Known solutions below $1e-60$; controls above $1e-6$	The exact result survives independent numeric checkpoints.
Rotation-context cache regression	$\rho^{**2} * \exp(-2 * z)$ valid at $\Omega=0$, invalid at $\Omega=1$	The pde-engine cache key must include rotation context.
Rotating determinant	$2048 * \rho^{**9} * \exp(-12 * z)$	The rotating failure is a symbolic fact, not only a cache artifact.

3. Engine process-control candidates admitted by the measurement

The PR now includes a second artifact, generated by `tools/export_force_free_novel_discoveries.py`. It runs the bounded pde-engine search, mines rows that are valid but not registered as one of the seven known force-free solutions, and then independently rebuilds the determinant instead of trusting the validator cache. The claim is scoped narrowly: these are process-control candidates, engine-novel relative to the repository registry, not claimed as literature-first discoveries.

Engine-discovered expression	Class	Independent witness
$\rho + z$	linear two-coordinate foliation	$\det M = 0$
ρ/z	scale-invariant ratio foliation	$\det M = 0$
$\rho^{**2} + z^{**2}$	quadratic radius foliation	$\det M = 0$
$\rho^{**2} + z$	parabolic polynomial foliation	$\det M = 0$
$-\rho^{**2} + z^{**2} + 1$	hyperbolic quadratic foliation	$\det M = 0$
$\rho / (1 - z)$	rational geometric-sum foliation	$\det M = 0$

The machine-readable version is `docs/force-free-novel-discoveries.json`; the review-friendly version is `docs/force-free-novel-discoveries.md`.

The full process record and next research target are documented in `docs/discovery-process-and-next-target.md`. The real paper target is not this force-free registry problem; it is a Kerr force-free Grad-Shafranov/extreme-Kerr target where the literature does not already provide an exact analytic solution meeting the same criteria.

4. Kerr paper-target gate

The next target gate is now a concrete artifact, not only a paragraph: `docs/kerr-paper-target-gate.json` and `docs/kerr-paper-target-gate.md`. It ran the current `kerr_magnetosphere` engine harness at depth 2, recorded 306 generated rows and 306 completed validations, then scanned all 306 generated expressions, 192 one-term correction candidates, 264 two-term correction candidates, and four probes against the full gate. It admitted 0 paper candidates.

This is the right failure mode. The current repository target is only a linear surrogate for generation, while paper admission is now decided by the closed split-monopole nonlinear residual from Mahlmann et al. Eq. `GSLightCylinder`, rewritten with $x = \cos(\theta)$. The strict gate requires dependence on r , x , and a ; finite denominators on rational safe points; a small-spin anchor to $1 - x$ or x ; rejection of those exact anchors as known; and exact full-residual zero after the point checks. No row meets all of those criteria, so the artifact status is `no_candidate_yet`.

The targeted correction grammar is deliberately bounded: $\Psi = 1 - x + a^2 c \text{basis}(r, x)$, with c in $(-1, -0.5, 0.5, 1)$, eight angular factors, and six radial factors. That Cartesian product generates 192 correction candidates. The two-term screen uses a 12-function sub-basis and sign coefficients to generate 264 more candidates. All 456 finite-spin correction candidates pass the cheap strict prechecks before the full residual, and all 456 fail exact-zero residual. The follow-up metric-resummed screen adds 112 Kerr-denominator candidates using `Sigma`, `Delta`, and the light-cylinder metric block; all 112 pass strict prechecks and all 112 fail exact-zero residual.

The stricter screen is not a larger fixed-coefficient search. It solves the leading a^2 residual coefficient for the full one-term basis: $\Psi = 1 - x + a^2 \sum_i c_i \text{basis}_i(r, x)$ with $M = 1$. That produces 66 polynomial equations in 48 unknown coefficients; the matrix has shape $[66, 48]$, and `linsolve` returns `EmptySet`. So even before the full nonlinear residual, this ansatz family has no leading-order correction.

The next calibration is positive but deliberately perturbative. The gate encodes the Blandford-Znajek split-monopole slow-rotation correction from Tanabe-Nagataki and Pan-Yu: $\Psi = 1 - x + a^2 x^*(1 - x^2) * R(r)$, where $R(r)$ contains the literature logarithm and dilogarithm radial terms. After applying $\text{polylog}(1, z) = -\log(1 - z)$, the coefficient of a^2 in the closed residual simplifies exactly to 0. This proves the gate recognizes the known $O(a^2)$ perturbative anchor, while still keeping it out of the finite-spin exact candidate set.

The JSON artifact now includes a criteria-status matrix. It marks the literature target, bounded engine generation, strict prechecks, nonlinear residual screen, finite-spin correction screens, leading-order coefficient solve, and literature perturbative anchor calibration as implemented. It deliberately marks broader equivalence filters and global horizon/axis/light-surface regularity as not sufficient for a positive paper claim. That distinction matters: this is a target packet and negative gate for a no-known-exact-solution problem, not a positive exact Kerr solution.

The literature motivation is explicitly separated from the current surrogate. Mahlmann et al. frame static, axisymmetric, force-free Kerr magnetospheres around the relativistic Grad-Shafranov equation and numerical solution methods. Camilloni et al. state that for extreme Kerr there is no known exact analytic stationary, axisymmetric, magnetically dominated force-free solution. The gate is designed for that claim family, not for more examples from the Compere force-free foliation registry.

5. The pde-engine patch forced by the measurement

The old validator cache hashed only the expression string. That was wrong: a non-rotating result could be replayed in a rotating context. The patch therefore includes the expression, schema, `Omega`, `check_regularity`, `fast_point_only`, and `use_lean` in the hash. The patch also removes the old fast-point placeholder derivative shortcut. Fast point mode still avoids the expensive full path, but it now builds the exact symbolic determinant before stopping at the exact point check.

The parallel reproduction worker had a separate bug: spawned children tried to import `physics_agent.problems`, which is not the package namespace in this repository. The patch switches the child import to `problems.load_problem` and adds a local fallback import for `PreciseFoliationValidator`.

6. Reproduction commands

```
python3 -m py_compile    general_method_paper_reproduction.py
problems/force_free/validator.py    tests/test_force_free_validator_cache.py
tools/export_force_free_novel_discoveries.py    tools/export_kerr_paper_target_gate.py
tests/test_force_free_novel_discoveries.py    tests/test_kerr_paper_target_gate.py

python3 -m unittest    tests/test_force_free_validator_cache.py
tests/test_force_free_novel_discoveries.py    tests/test_kerr_paper_target_gate.py

python3 tools/export_force_free_novel_discoveries.py    --run-engine --max-depth 2 --
validators 1    --timeout-s 90 --validation-timeout-s 3

python3 tools/export_kerr_paper_target_gate.py    --run-engine --max-depth 2 --validators 1
--timeout-s 90 --validation-timeout-s 3
```

The Lagra-side commands that produced the measurements were:

```
python3 experiments/proof-search/pde_engine_force_free_point_gate.py    --pde-engine-root
/Users/p/dev/pde-engine

python3 experiments/proof-search/pde_engine_reproduction_health_gate.py    --pde-engine-root
/Users/p/dev/pde-engine    --timeout-seconds 20

python3 experiments/proof-search/render_yukawa_finite_range_paper.py
python3 experiments/proof-search/publication_readiness_audit.py
python3 experiments/proof-search/proof_search_integrity_gate.py
```

7. References

[1] Geoffrey Compere, Samuel E. Gralla, Alexandru Lupasca, *Force-Free Foliations*, Phys. Rev. D 94, 124012 (2016), arXiv:1606.06727, DOI:10.1103/PhysRevD.94.124012. The paper formulates force-free electrodynamics in terms of field-line foliations; in the stationary axisymmetric case the object used here is a foliation of the half-plane.

Links: <https://arxiv.org/abs/1606.06727> and <https://doi.org/10.1103/PhysRevD.94.124012>.

[2] J. F. Mahlmann, P. Cerda-Duran, M. A. Aloy et al., *Numerically solving the relativistic Grad-Shafranov equation in Kerr spacetimes: Numerical techniques*, MNRAS 477, 3927-3946 (2018), arXiv:1802.00815, DOI:10.1093/mnras/sty858.

[3] F. Camilloni, G. Grignani, T. Harmark, R. Oliveri, M. Orselli, *Moving away from the Near-Horizon Attractor of the Extreme Kerr Force-Free Magnetosphere*, JCAP 10, 048 (2020), arXiv:2007.15665, DOI:10.1088/1475-7516/2020/10/048.

[4] K. Tanabe and S. Nagataki, *Higher Order Terms of Kerr Parameter for Blandford-Znajek Monopole Solution*, arXiv:0802.0908.

[5] Z. Pan and C. Yu, *Fourth-order split monopole perturbation solutions to the Blandford-Znajek mechanism*, arXiv:1503.05248.

8. Boundary

The adapter is a positive exact-symbolic and pointwise Lagra measurement of the pde-engine force-free boundary. It is not a Lean theorem. It is not a full pde-engine paper reproduction. The bounded depth-2 pde-engine smoke is now process-clean, and it yields six independently rechecked process-control

candidates not identical to the registered known-solution functions. The full reproduction boundary remains: the bounded pass only finds two known vertical canonical forms and does not recover the seven known Compere solutions. The Kerr gate is a separate `no_candidate_yet` artifact: it prevents the current linear surrogate from being promoted into the next paper target, appends targeted finite-spin corrections, and checks the closed split-monopole nonlinear residual directly. Its leading-order coefficient solve also returns `EmptySet` for the 48-function one-term ansatz. Its literature anchor screen does recognize the known $\mathcal{O}(a^{**2})$ Blandford-Znajek perturbative correction.